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ENBC 332

### Homework 4 Discussion and Outputs

#### Problem 1 Part C:

The sign of covariance indicates the direction of the linear relationship between the two variables of age and charges. The covariance I calculated for the dataset that I originally used in homework 2 was 50874.8 which indicates that they move in the same direction linearly. However, if the covariance was found to be negative, it would indicate that both variables are moving in the opposite direction of each other.

Pearson's correlation coefficient is a method of measuring the linear correlation between two sets of variables. More specifically, the formula used derives a number between negative one to positive one where a more negative value returned suggests that there is a negative linear relationship between both variables while a positive value would represent vice versa. The magnitude of the coefficient is also an indicator of how strong that relationship between both variables is. More specifically, the correlation is perfect if the coefficient is between 1 and 0.9, strong between 0.9 and 0.5, weak between 0.5 and 0.1, and uncorrelated below that.

For the dataset used from homework 2, I found that Pearson's correlation coefficient was approximately 0.2990082 which suggests a weak positive linear relationship between both variables.

As shown, the difference between Pearson's correlation coefficient and covariance is that the coefficient can only show how strong the linear relationship between two variables is positive or negative. Covariance is different where it can only provide information on whether the two trends given by both variables move in the same direction or in the opposite direction of each other.

Outputs:

Problem 1: